

Birzeit University
Discrete Math (Comp 233) First Exam
Section 9 — Answer Key & Marking Scheme
Second Semester 2025/2026

Question One [10 Marks] — MCQs (2 marks each)

1.1 If $n = 5k + 2$, what is $n^2 \bmod 5$?

Answer: (c) 4

$$n^2 = (5k + 2)^2 = 25k^2 + 20k + 4 = 5(5k^2 + 4k) + 4.$$

$$\text{So } n^2 \bmod 5 = 4.$$

1.2 Largest power of 6 dividing $30^5 \cdot 40^3$?

Answer: (b) 6^5

$$30 = 2 \cdot 3 \cdot 5, \text{ so } 30^5 = 2^5 \cdot 3^5 \cdot 5^5.$$

$$40 = 2^3 \cdot 5, \text{ so } 40^3 = 2^9 \cdot 5^3.$$

Product = $2^{14} \cdot 3^5 \cdot 5^8$. Since $6 = 2 \cdot 3$, the largest power of 6 dividing the product is $\min(14, 5) = 5$, giving 6^5 .

1.3 $m \bmod 6 = 3$ and $n \bmod 6 = 4$. What is $(m + n) \bmod 6$?

Answer: (b) 1

$$m = 6a + 3 \text{ and } n = 6b + 4 \text{ for some integers } a, b.$$

$$m + n = 6a + 6b + 7 = 6(a + b + 1) + 1.$$

$$\text{So } (m + n) \bmod 6 = 1.$$

1.4 Which is NOT equivalent to: $\forall$ integers n , if $4|n$ then $2|n$?

Answer: (d) Some integer divisible by 4 is divisible by 2

(a) is a direct verbal translation of the $\forall$ conditional — equivalent.

(b) negates the existence of a counterexample to the conditional — equivalent.

(c) “sufficient” matches the rule “ p sufficient for q means $p \rightarrow q$ ” — equivalent.

(d) uses $\exists$ (“some”), not $\forall$ — strictly weaker, so **not equivalent**.

1.5 $\exists$ a real number x such that $x^2 + 1 = 0$ — which best describes this?

Answer: (b) The statement is false; for every real x , $x^2 \geq 0$, so $x^2 + 1 \geq 1 > 0$.

For any real number x , $x^2 \geq 0$, so $x^2 + 1 \geq 1 > 0$. No real number satisfies the equation, so the existence claim is false.

Options (a) and (c) test simple substitutions: $0^2 + 1 = 1 \neq 0$ and $(-1)^2 + 1 = 2 \neq 0$.

Question Two [10 Marks] — Direct Proofs

A) [4 Marks] Prove that for every odd integer n , $n^2 + 3$ is divisible by 4.

Proof: Suppose n is any odd integer. [We must show $4 \mid (n^2 + 3)$.]

By the definition of odd, $n = 2k + 1$ for some integer k .

$$\text{Then } n^2 + 3 = (2k + 1)^2 + 3 = 4k^2 + 4k + 1 + 3 = 4k^2 + 4k + 4 = 4(k^2 + k + 1).$$

Let $m = k^2 + k + 1$; then m is an integer (sums and products of integers).

So $n^2 + 3 = 4m$, and by the definition of divisibility, $4 \mid (n^2 + 3)$. ■

Marking — Q2.A [4 marks]:

Marks	Criterion
1	Correct setup: introduce k via definition of odd ($n = 2k + 1$).
1	Correct algebraic expansion: $(2k + 1)^2 + 3 = 4k^2 + 4k + 4$.
1	Factoring out 4 to get $4(k^2 + k + 1)$.
1	Clean conclusion citing the definition of divisibility.

B) [3 Marks] Prove that for all integers a, b, c : if $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof: Suppose a, b, c are integers with $a \mid b$ and $b \mid c$. [We must show $a \mid c$.]

By the definition of divisibility, $b = ar$ for some integer r , and $c = bs$ for some integer s .

Substituting: $c = bs = (ar)s = a(rs)$.

Let $k = rs$; then k is an integer (product of integers).

So $c = ak$, and by definition $a \mid c$. ■

Marking — Q2.B [3 marks]:

Marks	Criterion
1	Correct application of the divisibility definition to both givens: $b = ar$ and $c = bs$.
1	Correct substitution to express c as $a(rs)$.
1	Clean conclusion: rs is an integer, hence $a \mid c$.

C) [3 Marks] Use QR with $d = 3$ to prove $n^2 + 2n$ has the form $3k$ or $3k + 2$.

Proof: Let n be any integer. By the quotient-remainder theorem with $d = 3$, exactly one of the following holds for some integer q :

$n = 3q$, $n = 3q + 1$, or $n = 3q + 2$.

Case 1 ($n = 3q$): $n^2 + 2n = 9q^2 + 6q = 3(3q^2 + 2q)$. Form $3k$ with $k = 3q^2 + 2q$.

Case 2 ($n = 3q + 1$): $n^2 + 2n = (3q + 1)^2 + 2(3q + 1) = 9q^2 + 6q + 1 + 6q + 2 = 9q^2 + 12q + 3 = 3(3q^2 + 4q + 1)$. Form $3k$.

Case 3 ($n = 3q + 2$): $n^2 + 2n = (3q + 2)^2 + 2(3q + 2) = 9q^2 + 12q + 4 + 6q + 4 = 9q^2 + 18q + 8 = 3(3q^2 + 6q + 2) + 2$. Form $3k + 2$.

In all three cases, $n^2 + 2n$ has the form $3k$ or $3k + 2$. ■

Marking — Q2.C [3 marks]:

Marks	Criterion
1	Correct setup applying QR with $d = 3$ to split n into three cases.
1	Correct algebraic computation of $n^2 + 2n$ in all three cases.

Marks	Criterion
1	Correct identification of the target form ($3k$ or $3k + 2$) for each case and clean conclusion.

Question Three [10 Marks]

A) [4 Marks total] Disprove “both odd $\rightarrow m + n$ is odd”, state correct conclusion, prove it.

(a) Counterexample [1 mark]:

Take $m = 1$ and $n = 3$. Both are odd, but $m + n = 4$, which is **even**, not odd. So the statement is false.

Acceptable variants: any specific pair of odd integers whose sum is even, e.g. (1, 1), (3, 5), (-1, 7).

(b) Correct conclusion and proof [3 marks]:

Correct conclusion: For all integers m and n , if m and n are both odd, then $m + n$ is **even**.

Proof: Suppose m and n are integers, both odd. By definition of odd, $m = 2j + 1$ and $n = 2k + 1$ for some integers j, k .

$$m + n = (2j + 1) + (2k + 1) = 2j + 2k + 2 = 2(j + k + 1).$$

Let $q = j + k + 1$; then q is an integer.

So $m + n = 2q$, and by definition $m + n$ is even. ■

Marking — Q3.A [4 marks total]:

Marks	Criterion
1	Valid counterexample: any specific pair of odd integers whose sum is even.
1	Correct statement of the corrected conclusion (“ $m + n$ is even ” when both odd).
1	Correct setup using definition of odd ($m = 2j + 1$, $n = 2k + 1$).
1	Completing the proof: show $m + n = 2(\text{integer})$, conclude even.

B) [6 Marks] Prove that for all rational r and s , $r^2 + s^2$ is rational.

Proof: Suppose r and s are rational numbers. [We must show $r^2 + s^2$ is rational.]

By the definition of rational, $r = a/b$ and $s = c/d$ for some integers a, b, c, d with $b \neq 0$ and $d \neq 0$.

Then $r^2 = a^2/b^2$ and $s^2 = c^2/d^2$.

$$r^2 + s^2 = a^2/b^2 + c^2/d^2 = (a^2d^2 + b^2c^2) / (b^2d^2).$$

The numerator $a^2d^2 + b^2c^2$ is an integer (sums and products of integers are integers).

The denominator b^2d^2 is an integer, and is nonzero since $b \neq 0$ and $d \neq 0$ (so $b^2d^2 \neq 0$).

Hence $r^2 + s^2$ is a quotient of integers with nonzero denominator, so by definition it is rational. ■

Marking — Q3.B [6 marks]:

Marks	Criterion
1	Setup using definition of rational: $r = a/b$, $s = c/d$, $b, d \neq 0$.

Marks	Criterion
1	Correct squaring: $r^2 = a^2/b^2$ and $s^2 = c^2/d^2$.
1	Correct combination into a single fraction with common denominator b^2d^2 .
1	Justifying the numerator is an integer.
1	Justifying the denominator is a nonzero integer (this point is often missed).
1	Clean conclusion citing the definition of rational.

Question Four [10 Marks]

A) [6 Marks total] Predicate logic — students (S), courses (C), $E(s, c)$.

(a) [3 marks] Symbolic form of: “There is a course that every student is enrolled in.”

Answer: $\exists c \in C$ such that $\forall s \in S, E(s, c)$.

Structure: $\exists \forall$ — the same course c must be enrolled in by every student s .

Common error: writing $\forall s \in S, \exists c \in C$ such that $E(s, c)$. This says every student is enrolled in **some** course (possibly different ones), which is a different, weaker claim.

Marking — Q4.A.a [3 marks]:

Marks	Criterion
1	Correct outer quantifier $\exists c \in C$.
1	Correct inner quantifier $\forall s \in S$, with proper scope.
1	Correct predicate $E(s, c)$ with arguments in the right order.

(b) [3 marks] Negation of $\forall s \in S, \exists c \in C$ such that $E(s, c)$.

Answer: $\exists s \in S$ such that $\forall c \in C, \sim E(s, c)$.

In English: “There is a student who is not enrolled in any course.”

Working: $\sim(\forall s, \exists c, E(s, c)) \equiv \exists s, \sim(\exists c, E(s, c)) \equiv \exists s$ such that $\forall c, \sim E(s, c)$.

Marking — Q4.A.b [3 marks]:

Marks	Criterion
1	Correct conversion of the outer $\forall$ to $\exists$.
1	Correct conversion of the inner $\exists$ to $\forall$.
1	Correct negation of the predicate $E(s, c)$ to $\sim E(s, c)$.

B) [4 Marks] Truth table for: $p \vee q, p \rightarrow r, q \rightarrow r, \therefore r$.

Truth table (T = true, F = false):

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	r	
T	T	T	T	T	T	T	← critical ✓
T	T	F	T	F	F	F	

T	F	T	T	T	T	T	← critical ✓
T	F	F	T	F	T	F	
F	T	T	T	T	T	T	← critical ✓
F	T	F	T	T	F	F	
F	F	T	F	T	T	T	
F	F	F	F	T	T	F	

Premise columns: the columns under $p \vee q$, under $p \rightarrow r$, and under $q \rightarrow r$.

Conclusion column: the column under r .

Answer: The argument is **valid**.

There are three critical rows where all three premises are true: $(p, q, r) = (T, T, T)$, (T, F, T) , and (F, T, T) .

In every critical row, the conclusion r is also true. Since there is no row where all premises are true but the conclusion is false, the argument form is **valid**. (This is the classic “proof by cases” inference rule.)

Marking — Q4.B [4 marks]:

Marks	Criterion
1	Complete and correct 8-row truth table with all premise and conclusion columns computed correctly.
1	Correct identification of which columns are premises ($p \vee q, p \rightarrow r, q \rightarrow r$) and which is the conclusion (r).
1	Correct identification of the three critical rows where all premises are true.
1	Correct validity verdict (valid) with explanation referencing the critical rows.

Marks Summary

Question One: 5 MCQs $\times$ 2 marks = 10 marks

Question Two: 4 + 3 + 3 = 10 marks

Question Three: 4 + 6 = 10 marks

Question Four: (3 + 3) + 4 = 10 marks

Total: 40 marks

General marking principles:

- Accept any logically equivalent reformulation of the symbolic answers in Q4.A.
- For proof questions, the marking breakdown is sequential — students lose individual marks for missing or incorrect steps rather than the whole question.
- Award full credit even if the student uses slightly different variable names (e.g. j, k, p, q , etc.) provided the logic is correct.
- In Q3.A counterexample, any specific pair of odd integers with even sum is acceptable. No partial credit for a generic claim without a specific witness.
- In Q4.B truth table, accept alternative correct column orderings provided premise and conclusion columns are clearly identified.